2027 Collegiate Rocket Launch Competition Handbook



Reflection of Rocket During Ascent
Photo Credit: Carthage College
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Table of Contents

Competition Objective 4
- Competition Engineering Parameters 4
- Judging Categories 5
- Competition Calendar 2026-2027 6
- CRL 2027 Flight Profile Example 8
- Competition Parameters Table 9

HPR Safety and Flight Overview 11
- Virtual Tech Inspection – Tripoli Wisconsin 11
- Overview of Safety Regulations 11
- High-power Rocketry Safety Code 13
- Safe Launch Practices 15
- Preflight Safety Inspection 17
- Post-flight Check-in 17

Report Deliverables 18
- Design Review Reports Objective 18
- Report Format 18
- Report Evaluation Criteria 18
- Report Scoring Formula 18

Competition Milestones 19
- Preliminary Design Review (PDR) 19
- Critical Design Review (CDR) 20
- Flight Readiness Review (FRR) 21
- Competition Flight 23
- Post-Flight Launch Assessment Review (PLAR) 24

Project Management 27

Educational Outreach 28

Appendix 29
- Appendix A-1 PDR Report Evaluation Example 29
- Appendix A-2 CDR Report Evaluation Example 30
- Appendix A-3 FRR Report Evaluation Example 31
- Appendix A-4 PLAR Report Evaluation Example 32
- Appendix A-5 Outreach Form 33

Competition Objective

The Wisconsin Space Grant Consortium's (WSGC) Collegiate Rocket Competition is intended to supply teams of affiliated university students with the opportunity to demonstrate engineering and design skills through direct application. It allows the teams to conceive, design, fabricate, and compete with high-power rockets. The restrictions on rocket motors and dimensions are limited so that knowledge, creativity and imagination of the students are challenged. The end result is a great aerospace experience for students that would not otherwise be available in the region.

Competition Engineering Parameters

Student teams will design, build, and fly a high-power, single-stage boosted-dart rocket consisting of a powered booster and an unpowered dart. Following motor burnout, the dart must separate under its own inertia from the booster and continue in a stable ascent flight to apogee. Both sections must descend under parachute and be safely recovered and return in a condition deemed flyable by the field Range Safety Officer.

The primary engineering objective of this year's competition is to successfully maximize the distance between the booster and the dart. In addition, teams must accurately predict the altitude of the dart at apogee and the booster's maximum altitude. The dart may not exceed the 4,000-foot Above Ground Level (AGL) ceiling. Teams must record sufficient on-board flight data to verify the booster's altitude, the dart's altitude, the separation event, and the performance of both sections throughout the flight.

The booster selection must have an outside diameter between 2 and 4 inches. The dart may be any diameter, provided that no portion (including fins) of the dart is wider than the booster. Teams are responsible for designing an aerodynamically stable configuration that maintains structural integrity during launch, separation, free flight, recovery, and landing.

The booster and dart must separate using a passive release system. Locking devices, black-powder separation charges, and other pyrotechnic separation methods are prohibited. Teams must demonstrate through testing and engineering analysis that the selected separation method will allow the booster and dart to disengage reliably without creating an unsafe flight condition.

Both sections of the rocket must include a recovery system. The booster portion may use a single-deployment recovery system utilizing motor ejection, but the dart section must be configured for dual-deployment recovery. A streamer is allowed at apogee in place of a drogue on the dart, followed by a main chute at a minimum of 300 ft AGL.

Teams must use one of the following approved rocket motors, provided by WSGC:

- AeroTech DMS 38mm H550ST Super Thunder
- AeroTech RMS 38mm I435T Blue Thunder
- AeroTech RMS 38mm I600 Redline

Each team rocket must utilize their own operational RRC3+ altimeter in the dart for flight-data collection and recovery-system control. All previous data must be deleted in the RRC3+ before competition flights. Teams may use additional avionics, altimeters, GPS or radio-frequency tracking systems, and audible locators as needed. Any additional electronic systems must not interfere with the required RRC3+ altimeter or other on-board equipment. An operational on-board camera(s) is(are) required to document the separation event and the flights of the booster and dart. A video recording device must be incorporated in the rocket dart, collecting down-looking on-board video. Teams may determine the number, location, and orientation of the cameras based on how they believe the

flight can be most effectively documented. Camera footage may be used to supplement, but not replace, the required on-board flight data.

All structural components and materials must be obtained from reputable high-power rocketry vendors or be supported by an engineering analysis demonstrating their suitability for the anticipated flight loads. Any custom-fabricated or nonstandard structural components must be submitted to the CRL Technical Team for review and approval.

The winner of the flight portion of the competition will be the team whose rocket completes a safe and successful flight while achieving the greatest verified altitude difference between the booster and dart. Final flight scoring will also consider successful mechanical separation, good quality camera operation, predicted altitudes of the booster and dart sections, compliance with the 4,000-foot AGL ceiling, reliable recovery of both sections, and the completeness and accuracy of the submitted flight data.

Judging Categories

Teams will be judged on the demonstration of their knowledge, the performance of their design, their ability to communicate effectively, and the workmanship of the fabrication. This will be accomplished in five parts: Design Review reports, Flight Readiness Review oral presentation, safe and successful competition flight (factoring in predicted vs. actual apogees and separation), project management, and Educational Outreach activity.

The total score for each student team will be based on the following parameters:

Milestone Reports (PDR, CDR, FRR, PLAR)	(Written)	25
Flight Readiness Review Presentation	(Oral)	10
Competition Flight	(Flight)	50
Project Management	(Performance)	05
Educational Outreach	(Form)	10
	Total	**100**

Competition Calendar 2026-2027

All virtual meetings are held through Zoom: https://zoom.us/j/92786761125

Sep 21, 2026	**Informational Virtual Meeting @ 7:00 pm** - Zoom (Mon)
Oct 19, 2026	**Notice of Intent to Compete Due** (Mon)
Oct 19, 2026	**Launch 2 Learn (L2L) Applications Due** (Mon)
Oct 26, 2026	**CRL27 Competition Award Announcements (Mon)**
Nov 2, 2026	**Kick-Off Virtual Meeting @ 7:00 pm** - Zoom (Mon)
Nov 2, 2026	**Award Acceptance Material Due** (Mon)
Nov 5, 2026	**L2L Kit Reveal @ 6:00 pm - Zoom (Thu)**
Nov 13-14, 2026	**L2L Virtual Workshop:** Build Rocket to qualify for Level 1 Certification - Zoom (Fri PM, Sat)
Dec 7, 2026	**Preliminary Design Report Milestone** (Mon), Deliverables Due:
	1. **PDR Report*** 2. **Preliminary Budget*** 3. **RockSim** (must be full RockSim, no OpenRocket) 4. **Demo Flight Recording** ○ Upload video on Facebook ○ Upload link on Team Lead grant management page on WSGC Carthage site
Dec 10-11, 2026	**PDR Virtual Tag-Up: 6:00-8:00 pm**, each team by appointment **-** Zoom (Thu-Fri)
	● Discuss final motor selection
Jan 25, 2027	**Mandatory Competition Update and Q&A Virtual Meeting @ 7:00 pm** - Zoom (Mon)
Feb 8, 2027	**Critical Design Report Milestone** (Mon), Deliverables Due:
	1. **CDR Report*** 2. Updated **RockSim** (must be full RockSim, no OpenRocket) 3. **Final Team Roster** 4. **Team Photo** 5. **Team Bio**
Feb 17-18, 2027	**CDR Virtual Review 6:00-9:00 pm**, each team by appointment - Zoom (Wed-Thu)
Feb 19, 2027	**Deadline to Reserve Hotel Rooms in WSGC Block for Launch Weekend** (Fri) ● *Hampton Inn & Suites, Kenosha, WI Booking Link (Available Soon)*
Mar 1, 2027	**First Payout Deadline** (Mon)
	● Complete and email WSGC Expense Reimbursement Form (found in Resources) to the WSGC Program Office (wsgc.accounts@carthage.edu). Include original receipts. Allow 60 days for payment.
Mar 18 & 25, 2027	**Virtual Safety and Design Review Meetings**, each team by appointment - Zoom (Thu)
	● Mandatory meeting with 90% ready-to-fly rocket
Apr 5, 2027	**Flight Readiness Report Milestone** (Mon), Deliverables Due:
	1. **FRR Report*** 2. Updated **RockSim** (must be full RockSim, no OpenRocket) 3. **TRA/NAR Team Membership** forms 4. **Education Outreach* Form Deadline**
Apr 19, 2027	**FRR Oral Presentation PowerPoint* Deadline** (Mon)
	● Uploaded to WSGC Grant Management Page (no modifications after submission without late penalty)

Apr 23, 2027	**FRR Oral Presentation at Carthage College 6:00-9:00 pm** (Fri)
	• In-person 6-8 minute PowerPoint presentation discussing team's rocket
Apr 24, 2027	**Launch Competition: Complete High-Power Rocket Launch 8:00 am-3:00 pm** (Sat)
	• Richard Bong Recreational Area in Kansasville, WI
Apr 25, 2027	**Launch Competition: Rain Date/Certification Flights 8:00 am-3:00 pm** (Sun)
	• Richard Bong Recreational Area in Kansasville, WI
May 10, 2027	**Post-Launch Assessment Report Milestone** (Mon)
	1. **PLAR Report*** 2. **Final Payout Deadline** 3. Complete and email WSGC Expense Reimbursement Form (found in Resources) to the WSGC Program Office (wsgc.accounts@carthage.edu). Include original receipts. Allow 60 days for payment. 4. **Project Photos**
June 15, 2027	**CRL27 Award Announcement**
Sum-Fall 2027	**Sierra Space Tour (Date TBD)**
Aug 13, 2027	**Annual Wisconsin Space Conference (Fri)**
	• Teams placing 1st-3rd in the competition are required to present their results associated with this program at the 37th Annual Wisconsin Space Conference
Sep 15, 2027	**Wisconsin Space Conference Proceedings Paper**** (Wed)
	• Teams placing 1st-3rd in the competition are invited to submit a Proceedings Paper for the 37th Annual Wisconsin Space Conference online journal

*Submitted documents must be uploaded to the *grant management page, and **online journal by the Team Lead. Submissions received after 11:59 CDT/CST will be considered late with 20% penalty per any increment of each successive day.*

CRL 2027 Flight Profile Example

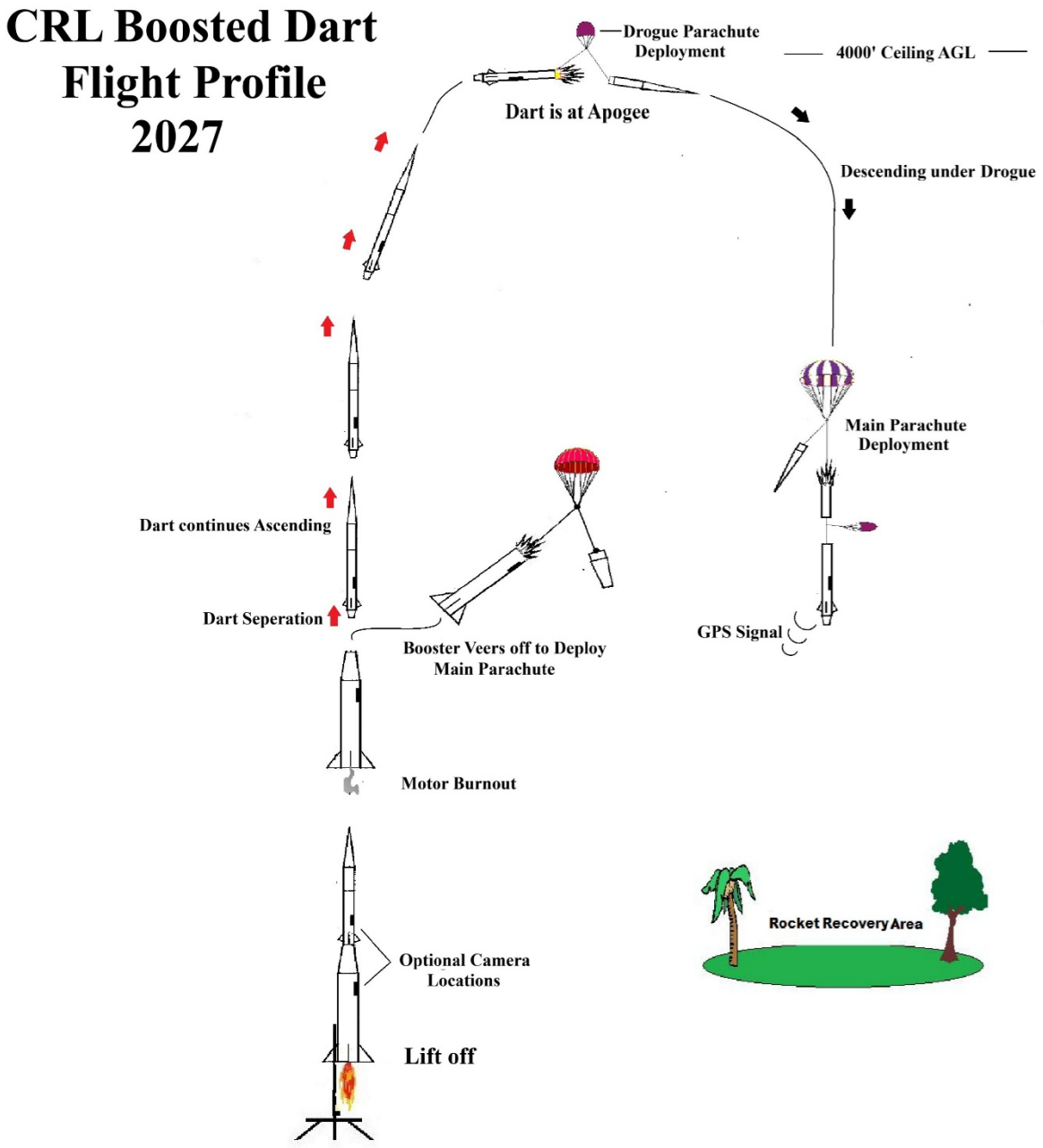


Figure 1: Flight Path of Rocket

Competition Parameters Table

Flight Mission	• Safe and Successful Flight • Include an avionics system that will ○ Monitor and record: ▪ Altitude ▪ Acceleration, to allow identification of boost phase as well as when deployment occurs • Avionics system may not be a single-unit Commercial Off-the-Shelf (COTS) system designed for rockets or Unmanned Aerial Systems (UAS, aka drones). Teams may integrate COTS components to engineer their avionics system.
Altitude Required	• Apogee does not exceed the 4,000 ft AGL ceiling
Target Altitude	• Maximum vertical separation between booster and dart apogees
On-Board Video	• A video recording device must be incorporated in the rocket dart, collecting down-looking on-board video.
Recovery	• Electronic, dual-deployment recovery deployment for dart ○ Two separate recovery deployment chambers are required ○ Delayed-open chute release devices are NOT allowed (aka Jolly Logic Chute Release) • Motor recovery deployment is required as backup for safety • Parachutes (required) ○ A streamer may be used in place of the drogue in the dart, a standard parachute is required for main deployment. • Electronic GPS/RF rocket location aid (required in dart, optional in booster)
Rocket Constraints	• Minimum 5:1 thrust to weight ratio • Each team must prepare a mounting location for the competition flight recorder within their rocket • Each team must indicate (mark and label) the Center of Pressure (CP) and Center of Gravity (CG) on the outside of the rocket airframe Include CP and CG for both the fully-loaded rocket and after the motor burnout
High-Power Rocketry Associations	• In order to be covered by insurance, **ALL** members of the competition team **MUST** be members in good standing of either: Tripoli Rocketry Association, www.tripoli.org or National Association of Rocketry, www.nar.org • While encouraged, it is not a requirement that team members be certified to a certain HPR level
NASA STEM Gateway	• ALL members of the competition team must have an updated profile on NASA STEM Gateway: https://stemgateway.nasa.gov
Low-power Rocket Demonstration Flight	• Each team must purchase, assemble, fly, and successfully recover a "low-power" rocket Pictures of the team at their launch site with the rocket, before and after the launch, along with video capturing the flight ○ Must be posted to the WSGC Facebook Page or LinkedIn when the budget is submitted Example "low-power" rockets: Estes Alpha or Quest Astra

Rocket Design and Safety Reviews	● Each team must work with WSGC to identify a Rocketry Mentor. The rocketry mentor must hold a current TRA or NAR High-Power Level 2 or 3 certification and is not a member of the team ● At a minimum, three (3) design reviews (can be virtual) must be held with the rocketry mentor ● Each team, with their rocket, must participate in the virtual Design and Safety Review meeting approximately one month before the competition launch ● Analysis of non-pre-qualified components must accompany the rocket at the Design and Safety Review ● Rockets must be 90% ready-to-fly. **Teams without rockets at this level of construction will not be allowed to continue in the competition** ● Each rocket must pass the Range Safety Officer's Inspection on the day of launch, before it will be allowed to fly
Educational Outreach	● Each team must share STEM-related information with a group or audience. Outreach efforts will be evaluated based on the audience reached, the type and scope of activities conducted, and the extent of ongoing engagement. Consideration will be given to whether the event was new or continuing, its duration and format, and any follow-up efforts or documented impact.
Safe and Successful Flight	● Launch followed by stable, near-vertical trajectory during ascent ● Rocket's electronic, dual-deployment recovery system functions successfully, deploying drogue and main parachutes (and motor backup deploys successfully) ● Apogee does not exceed the maximum altitude of 4,000 ft AGL ceiling ● Acquisition of downward-looking video from dart throughout flight ● Rocket booster and dart must be recovered safely and in a ready-to-fly condition

In addition to the budget support, WSGC will provide each team with the following equipment:

Required Competition Motor	● Motor must be selected from the following (all 38mm): o AeroTech DMS 38mm H550ST Super Thunder o AeroTech RMS 38mm I435T Blue Thunder o AeroTech RMS 38mm I600 Redline

HPR Safety and Flight Overview

The Federal Aviation Administration (FAA) (www.faa.gov) has specific laws governing the use of airspace. A demonstration of the understanding and intent to abide by the applicable federal laws (especially as related to the use of airspace at the launch sites and the use of combustible/flammable material), safety codes, guidelines, and procedures for building, testing, and flying large model rockets is crucial. The procedures and safety regulations of the TRA (http://www.tripoli.org/SafetyCode /) shall be used for flight design and operations. The NAR/TRA mentor and Safety Officer shall oversee launch operations and motor handling.

Virtual Tech Inspection – Tripoli Wisconsin

All teams are required to participate in a Virtual Tech Inspection approximately one month before Launch Weekend. The teams must be prepared to discuss the design of their rocket and its systems. In addition, the teams must display:

- Rockets must be 90% ready-to-fly
    - **Teams without rockets at this level of construction will not be allowed to continue in competition.**
- Bench test operation of release mechanism system
- Explanation of the avionics system
- Explanation of the dual-deployment recovery system in the dart and how the booster allows for motor deployment as a backup. Dual-deployment recovery of the booster section is optional
- A diagram of the rocket indicating the configuration of its main components
- Analysis of non-pre-qualified components must accompany the rocket
- RockSim flight simulation showing max altitude and launch guide velocity (min. 50 ft/s at 10 ft altitude)
- Deployment altimeter user manual
- Preflight Checklist
- Launchpad and Flight Arming checklist
    - Must include the altimeter's ready/standby tones
- Recovery/Post-flight Checklist
    - Must include procedure to "safe" deployment charges and payload

The team will be given a go/no-go by the WSGC Technical Advisor. The Technical Advisor must be satisfied with the state of build to proceed to competition weekend. The schedule will be posted at a later date.

Overview of Safety Regulations

High-power rocketry is federally regulated by the National Fire Protection Association (NFPA). National rocketry organizations, including the Tripoli Rocketry Association – TRA (http://www.tripoli.org) and the National Association of Rocketry – NAR (http://www.nar.org), also have safety guidelines and regulations to follow. The purpose of NFPA 1127, the Tripoli Safety Code, and the NAR Safety Code is to:

1. Provide safe and reliable motors, establish flight operations guidelines, and prevent injury.
2. Promote experimentation with rocket designs and payload/challenge systems.
3. Prevent beginning high-power hobbyists from making mistakes.

Detailed NFPA, TRA and NAR Safety Regulations may be found at the following links:

National Fire Protection Association
NFPA 1127 Code for High-power Rocketry
http://www.nfpa.org/1127

Tripoli Rocketry Association
Tripoli Code for High-power Rocketry
Safety Information - Tripoli Rocketry Association

National Association of Rocketry
NAR High-power Rocket Safety Code
https://www.nar.org/content.aspx?page_id=22&club_id=114127&module_id=669238

HPR Launch Sites
Contact a local NAR or Tripoli Club that has an FAA Waiver, a designated launch site, and established club launch dates where you can safely conduct test flights, etc.

The Federal Aviation Administration (FAA) regulates and classifies model rockets according to FAR 101 Subpart C, which is summarized in Table 1. NAR and Tripoli certification requirements and limitations can be seen in Table 2. See the FARs for more details.

Table 1: FAA Rocket Classification

Limitation	Class 1	Class 2
Rocket Weight	No more than 1500 grams	No limit
Motor Size Limit	No more than 125 grams	No more than 40960 N-sec total thrust
Altitude Limit	None – may be set by local agreement	FAA limited
Other	Clear of clouds	Must have 5 miles horizontal visibility, clouds less than 5/10ths coverage, FAA Waiver and NOTAM filed between sunrise and sunset

Table 2: High-Power Rocketry Certification Requirements

	Certification Required			
Motor Parameter	None	Level 1 HPR	Level 2 HPR	Level 3 HPR
Total Combined Impulse	320 N-sec (2xG Class)	640 N-sec (H, I Class)	5120 N-sec (J, K, L Class)	40960 N-sec (M,N,O Class)
Combined Propellant Mass	125 grams	No Limit		
Single Motor Impulse	160 N-sec	No Limit		
Single Motor Propellant Mass	62.5 grams	No Limit		
Single Motor Avg Thrust	80 N	No Limit		
Sparky Motors	Not Allowed	Allowed		
Total Rocket Mass	1500 grams	No Limit		
Field Distance Reqmts	Per Model Rocket Safety Code	Per HPR Safety Code		

High-power Rocketry Safety Code

1. **Certification.** I will only fly high-power rockets or possess high-power rocket motors that are within the scope of my user certification and required licensing.
2. **Materials.** I will use only lightweight materials such as paper, wood, rubber, plastic, fiberglass, or when necessary ductile metal, for the construction of my rocket.
3. **Motors.** I will use only certified, commercially made rocket motors, and will not tamper with these motors or use them for any purposes except those recommended by the manufacturer. I will not allow smoking, open flames, nor heat sources within 25 feet of these motors.
4. **Ignition System.** I will launch my rockets with an electrical launch system, and with electrical motor igniters that are installed in the motor only after my rocket is at the launchpad or in a designated prepping area. My launch system will have a safety interlock that is in series with the launch switch that is not installed until my rocket is ready for launch, and will use a launch switch that returns to the "off" position when released. The function of on-board energetics and firing circuits will be inhibited except when my rocket is in the launching position.
5. **Misfires.** If my rocket does not launch when I press the button of my electrical launch system, I will remove the launcher's safety interlock or disconnect its battery, and will wait 60 seconds after the last launch attempt before allowing anyone to approach the rocket.
6. **Launch Safety.** I will use a 5-second countdown before launch. I will ensure that a means is available to warn participants and spectators in the event of a problem. I will ensure that no person is closer to the launchpad than allowed by the accompanying Minimum Distance Table. When arming on-board energetics and firing circuits I will ensure that no person is at the pad except safety personnel and those required for arming and disarming operations. I will check the stability of my rocket before flight and will not fly it if it cannot be determined to be stable. When conducting a simultaneous launch of more than one high-power rocket I will observe the additional requirements of NFPA 1127.
7. **Launcher.** I will launch my rocket from a stable device that provides rigid guidance until the rocket has attained a speed that ensures a stable flight, and that is pointed to within 20 degrees of vertical. If the wind speed exceeds 5 miles per hour, I will use a launcher length that permits the rocket to attain a safe velocity before separation from the launcher. I will use a blast deflector to prevent the motor's exhaust from hitting the ground. I will ensure that dry grass is cleared around each launchpad in accordance with the accompanying Minimum Distance table, and will increase this distance by a factor of 1.5 and clear that area of all combustible material if the rocket motor being launched uses titanium sponge in the propellant.
8. **Size.** My rocket will not contain any combination of motors that total more than 40,960 N-sec (9208 pound-seconds) of total impulse. My rocket will not weigh more at liftoff than one-third of the certified average thrust of the high-power rocket motor(s) intended to be ignited at launch.
9. **Flight Safety.** I will not launch my rocket at targets, into clouds, near airplanes, nor on trajectories that take it directly over the heads of spectators or beyond the boundaries of the launch site, and will not put any flammable or explosive payload/challenge in my rocket. I will not launch my rockets if wind speeds exceed 20 miles per hour. I will comply with Federal Aviation Administration airspace regulations when flying, and will ensure that my rocket will not exceed any applicable altitude limit in effect at that launch site.
10. **Launch Site.** I will launch my rocket outdoors, in an open area where trees, power lines, occupied buildings, and persons not involved in the launch do not present a hazard, and that is at least as large on its smallest dimension as one-half of the maximum altitude to which rockets are allowed to be flown at that site or 1500 feet, whichever is greater, or 1000 feet for rockets with a combined total impulse of less

than 160 N-sec, a total liftoff weight of less than 1500 grams, and a maximum expected altitude of less than 610 meters (2000 feet).

11. **Launcher Location.** My launcher will be 1500 feet from any occupied building or from any public highway on which traffic flow exceeds 10 vehicles per hour, not including traffic flow related to the launch. It will also be no closer than the appropriate Minimum Personnel Distance from the accompanying table from any boundary of the launch site.
12. **Recovery System.** I will use a recovery system such as a parachute in my rocket so that all parts of my rocket return safely and undamaged and can be flown again, and I will use only flame-resistant or fireproof recovery system wadding in my rocket.
13. **Recovery Safety.** I will not attempt to recover my rocket from power lines, tall trees, or other dangerous places, fly it under conditions where it is likely to recover in spectator areas or outside the launch site, nor attempt to catch it as it approaches the ground.

Safe Launch Practices

I. **All Launches:**

A. Must comply with United States Code 1348, "Airspace Control and Facilities," Federal Aviation Act of 1958 and other applicable federal, state, and local laws, rules, regulations, statutes, and ordinances.

B. A person shall fly a rocket only if it has been inspected and approved for flight by the RSO. The flier shall provide documentation of the location of the Center of Pressure (CP) and the Center of Gravity (CG) of the high-power rocket to the RSO if the RSO requests the same.

C. The member shall provide proof of membership and certification status by presenting their membership card to the Launch Director or RSO upon request.

D. A rocket with a predicted altitude in excess of 50,000 feet AGL requires review and approval by the TRA Class 3 Committee.

E. Recovery

1. Fly a rocket only if it contains a recovery system that will return all parts of it safely to the ground so that it may be flown again.
2. Ensure that adequate protection is in place to prevent hot ejection gasses from causing burn damage to retaining cords, parachutes, and other vital components.
3. Do not attempt to catch a high-power rocket as it approaches the ground.
4. Do not attempt to retrieve a rocket from a power line or other place that would be hazardous to people attempting to recover it.

F. Payload/challenges

1. Do not install or incorporate in a high-power rocket a payload/challenge that is intended to be flammable, explosive or debris that can cause harm.
2. Do not fly a vertebrate animal in a high-power rocket.

G. Weight Limits

1. The maximum lift-off weight of a rocket shall not exceed one-third (1/3) of the average thrust on the motor(s) intended to be ignited at launch.

H. Launching Devices

1. Launch from a stable device that provides rigid guidance until the rocket has reached a speed adequate to ensure a safe flight path.
2. Incorporate a jet/blast deflector device if necessary to prevent the rocket motor exhaust from impinging directly on flammable materials.

I. Ignition Systems

1. Use an ignition system that is remotely controlled, electrically operated, and contains a launching switch that will return to "off" when released.
2. The ignition system shall contain a removable safety interlock device in series with the launch switch.
3. The launch system and igniter combination shall be designed, installed, and operated so the liftoff of the rocket shall occur as quickly as possible after actuation of the launch system. If the rocket is propelled by a cluster of rocket motors designed to be ignited simultaneously, install an ignition scheme that has either been previously tested or has a demonstrated capability of igniting all rocket motors intended for launch ignition within one second following ignition system activation.
4. A rocket motor shall not be ignited by a mercury switch or roller switch.

a) Install an ignition device in a high-power rocket motor only at the launchpad.

J. Launch Operations
   1. Do not launch with surface winds greater than 20 mph (32 km/h) or launch a rocket at an angle more than 20 degrees from vertical.
   2. Do not ignite and launch a high-power rocket horizontally, at a target, in a manner that is hazardous to aircraft, or so the rocket's flight path goes into clouds or beyond the boundaries of the flying field (launch site).
   3. A rocket shall be pointed away from the spectator area and other groups of people during and after installation of the ignition device(s).
   4. Firing circuits and on-board energetics shall be inhibited until the rocket is in the launching position.
   5. Firing circuits and on-board energetics shall be inhibited prior to removing the rocket from the launching position.
   6. When firing circuits for pyrotechnic components are armed, no person shall be allowed at the pad area except those required for safely arming/disarming.
   7. Do not approach a high-power rocket that has misfired until the RSO/LCO has given permission.
   8. Conduct a five second countdown prior to launch that is audible throughout the launching, spectator, and parking areas.
   9. All launches shall be within the Flyer's certification level, except those for certification attempts.
   10. The RSO/LCO may refuse to allow the launch or static testing of any rocket motor or rocket that he/she deems to be unsafe.

II. Commercial Launches
   A. Use only certified rocket motors.
   B. Do not dismantle, reload, or alter a disposable or expendable rocket motor, nor alter the components of a reloadable rocket motor or use the contents of a reloadable rocket motor reloading kit for a purpose other than that specified by the manufacturer in the rocket motor or reloading kit instructions.
   C. Do not install a rocket motor or combination of rocket motors that will produce more than 40,960 N-s of total impulse.
   D. Rockets with more than 2560 N-s of total impulse must use electronically actuated recovery mechanisms.
   E. When more than 10 model rockets are being launched simultaneously, the minimum spectator distance shall be set to 1.5 times the highest altitude expected to be reached by any of the rockets.
   F. When three or more rockets (at least one high-power) are launched simultaneously, the minimum distance for all involved rockets shall be the lesser of:
      1. Twice the complex distance for the total installed impulse. (Refer to V. Distance Tables)
      2. 2000 ft. (610 m)
      3. 1.5 times the highest altitude expected to be achieved by any of the rockets.
         (1) When more than one high-power rocket is being launched simultaneously, a minimum of 10 ft. (3m) shall exist between each rocket involved.

Table 3: Minimum Distance Table

Installed Total Impulse (Newton-Seconds)	Equivalent High Power Motor Type	Minimum Diameter of Cleared Area (ft.)	Minimum Personnel Distance (ft.)	Minimum Personnel Distance (Complex Rocket) (ft.)
0 – 320.00	H or smaller	50	100	200
320.01 – 640.00	I	50	100	200
640.01 – 1,280.00	J	50	100	200
1,280.01 – 2,560.00	K	75	200	300
2,560.01 – 5,120.00	L	100	300	500
5,120.01 – 10,240.00	M	125	500	1000
10,240.01 – 20,480.00	N	125	1000	1500
20,480.01 – 40,960.00	O	125	1500	2000

Preflight Safety Inspection

On flight competition day, all teams must have their rockets inspected before they are allowed to proceed to the launchpad. The teams must be prepared to discuss their rocket's design and its deployment systems. In addition, the teams must display:

- Team's rocket readied for launch
  - The Center of Gravity (CG), both on the launch pad and after motor burnout, and the Center of Pressure (CP) must be clearly marked on the rocket's exterior.
  - Preflight Checklist (showing that all steps have been completed up to launch).
  - Launchpad and Flight Arming checklist.
    Must include the altimeter's ready/standby tones.
  - Recovery/Post-flight Checklist.
    Must include procedure to "safe" deployment charges and payload.

Post-flight Check-in

Following the team's competition flight, the team must follow their Recovery/Post-flight Checklist to ensure a safe recovery. The team then proceeds to the recovery check-in with:

- The team's rocket
- Recovery/Post-flight Checklist
  - Must show that all steps in the recovery procedure were completed before approaching the check-in station.
- A flight performance report sheet will be filled out by a designated flight operations recorder. They will record the data on the sheet during and following the flight. Upon completion, a team member must sign their initials of acceptance before a copy will be released to the team.

Report Deliverables

Design Review Reports Objective

The concept of the design review reports is to evaluate the process and effort that the team put into the design of the rocket and how the engineering meets the intent of the competition. The rocket that illustrates the best use of design to meet the design goals and the best understanding of the design by the team members will score the highest. There are three (3) design reports and one (1) Post-Launch Assessment Report (PLAR) that must be submitted. The first is the Preliminary Design Review (PDR) report, which describes the design goals, constraints, proposed solution idea, preliminary project schedule, and proposed budget. The second report is the Critical Design Review (CDR) report that describes how the design was developed, the design of the solution, and predictions for its performance. The third report is the Flight Readiness Review (FRR) report, which examines how the design was altered following the CDR given the feedback from the Safety Review, as well as any payload performance testing and rocket flight tests. The PLAR examines the results of the flight and the performance of the selected method used to address the competition challenge.

Report Format

Each report must use single spacing and a single-sided page format. They must be in a font no smaller than 12 pt. The left margin must be no less than 1 inch and the remaining margins must be no less than 1 inch from the edge of the page. All pages (except for the cover page) must be numbered in the upper right-hand corner. Each section of the report must be clearly delineated with a heading. All section headings must appear in a table of contents. Reports must be submitted electronically in .pdf format.

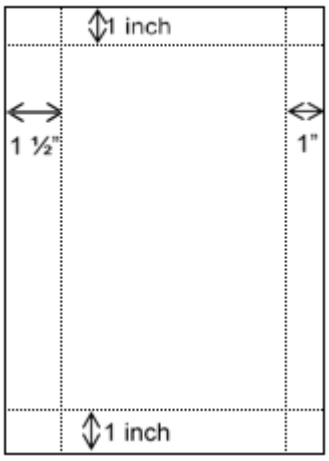


Figure 2: Design Report Page Layout

Report Evaluation Criteria

Reports and designs will be evaluated on content, organization, clarity, completeness, and professionalism of the material.

Report Scoring Formula

The scoring of each required review and analysis report is based on the average of the Design Report Judging forms for the respective report. Then the team's review report scores will be combined and scaled to meet the required portion of the competition total score.

Competition Milestones

Preliminary Design Review (PDR)

PDR Report

The PDR must be a short report using single-sided pages and must include, at a minimum:

- **Cover Page**
  - Team name
  - School
  - Team Members
  - Faculty Advisor
  - Rocketry Mentor, if known
- **Table of Contents**
- **Design Goals**
  - List the design goals for the rocket and payload
    - How is the rocket and payload performance measured by the competition?
    - How will the team ensure stable flight properties throughout the flight and during the separation of the dart?
    - Which will be designed first? Rocket, payload, or simultaneously?
    - What does the team define as most important?
- **Design Constraints**
  - List limits on design:
    - Limits imposed by the competition
    - Limits imposed by the motor options
    - Limits team has chosen to include in addition to those required by the competition
  - List limits on resources:
    - Budget
    - People
    - Skills/Tools
    - Time
- **Description of Preliminary Solution Idea**
  - Briefly describe the solution the team has chosen to advance
  - RockSim model results, motor, dimensions, CP, CG
  - Images or sketches of proposed rocket and payload, details
- **Proposed Schedule**
  - Identify competition deliverable dates
  - Identify competition meeting dates (include plan for Wisconsin Space Conference attendance)
  - Identify school important dates (school breaks, final exam days, vacations, etc.)
  - Show design, fabrication, documentation/reporting, subsystem and system test milestones
- **Proposed Budget**
  - All rocket components
  - All payload components
  - Travel expenses
- **Safety**
  - Designed for safe flight and recovery
  - Pre- and post-launch checklists

Critical Design Review (CDR)

CDR Report

The CDR Report must be no more than twenty-five (25) single-sided pages in length and must include, at a minimum:

- **Cover Page** *(not included in page total)*
- **Table of Contents** *(not included in page total)*
- **Executive Summary**
    - One or two paragraphs describing the rocket, payload, and predicted performance
    - Image of the rocket
- **Design Features of Payload**
    - Background information on possible methods for boosted dart and separation
    - Brief description of each possible method identified
    - Comparison of possible methods, including the decision process for evaluation and selection of the method employed
    - Design of an avionics system and any video recording system
    - Image of the payload
- **Design Features of Recovery Systems**
    - Selected altimeter for the recovery system deployment and how it is mounted
    - Dual-electronic deployment system (primary)
    - Motor deployment system (required backup)
    - Parachutes selected (a streamer may be used in place of a drogue chute in the dart)
    - Shock-cord and mountings
- **Accurate Diagram of Rocket**
    - Identifying the dimensioned locations for the:
        - Center of Pressure (CP) of the rocket
        - Center of Gravity (CG), denote location for fully loaded rocket (motor, payload, batteries, etc.) when on the launchpad and the expected location following motor burnout
- **Analysis of Anticipated Performance**
    - Including how each was estimated:
        - Liftoff thrust to weight ratio
        - Estimated launch rail guide velocity (speed at 10 ft altitude)
        - Estimated maximum altitude of rocket booster and dart
        - Estimated peak acceleration
        - Plot of Estimated Acceleration of rocket booster and dart vs. time (from start of flight)
        - Plot of Estimated Altitude of rocket booster and dart vs. time (from start of flight)
- **Construction of Rocket Airframe and Payload** (include photos)
- **Conclusion**
- **Photographs of Completed Rocket and Payload** *(not included in page total)*
- **Budget** *(not included in page total)*

Flight Readiness Review (FRR)

FRR Report

The FRR Report must be no longer than ten (10) single-sided pages in length and must include, at a minimum:

- **Cover Page** *(not included in page total)*
- **Table of Contents** *(not included in page total)*
- **Executive Summary**
  - One or two paragraphs summarizing adjustments made to the rocket, payload, and predicted performance following the Safety Review, payload performance tests, and any test flights
  - Image of the rocket and payload
- **Results of Payload System Performance Verifications**
  - Avionics system testing
  - Rocket mounted video system testing
- **Adjustment to Design of Payload**
  - Document adjustments made to the design of the payload during fabrication and payload performance tests
  - Volume (in^3) of main deployment chamber
  - Grams of black powder and area (mm^2) of air holes in main deployment chamber
  - Number and size of shear pins used for main deployment chamber
  - Image of the payload
- **Adjustments to Design of Rocket**
  - Document adjustments made to rocket design during fabrication and flight tests
  - Image of final rocket
- **Accurate Diagram of Rocket**
  - Identifying the dimensioned locations for the:
    - Center of Pressure (CP) of the rocket
    - Center of Gravity (CG), denote location for fully loaded rocket (motor, payload, batteries, etc.) when on the launchpad, and the expected location following motor burnout
- **Adjustment to Anticipated Performance**
  - Including how each was estimated:
    - Liftoff thrust to weight ratio
    - Estimated launch rail guide velocity (speed at 10 ft altitude)
    - Estimated maximum altitude of rocket
    - Estimated peak acceleration
    - Plot of Estimated Acceleration of rocket vs time (from start of flight)
    - Plot of Estimated Altitude vs time (from start of flight)
- **Conclusion**
- **Photographs of Completed Rocket and Payload** *(not included in page total)*
  - With CP and both CGs indicated on exterior of rocket airframe

FRR Presentation (Oral)

FRR Presentation Format

Team members will deliver the presentation to the judges in front of an audience. All team members who will deliver any part of the presentation, or who will respond to the judges' questions, must be in the podium area when the presentation starts and must be introduced to the judges. Team members who are part of this "presentation group" may answer the judges' questions even if they did not speak during the presentation itself.

Presentations are limited to a 6-8 minute PowerPoint presentation. The judges will stop any presentation exceeding ten minutes. The presentation itself will not be interrupted by questions. Immediately following the presentation, there will be a question-and-answer session of up to 2-4 minutes. Only judges may ask questions. Only team members who are part of the "presentation group" may answer the judges' questions. If time allows, there may be an opportunity to take additional questions from the audience. If questions are taken from the audience, a designated presentation official will determine if the question is appropriate, and if so, then allow the team to answer.

FRR Presentation Evaluation Criteria

Presentations will be evaluated on content, organization, visual aids, delivery, and the team's response to the judges' questions. Rockets will be evaluated for the quality of their fabrication. The scoring criteria are detailed in Appendix A-3 "FRR Report Evaluation Example". The criteria are applied only to the team's presentation itself. The team that makes the best presentation, regardless of the quality of their rocket, will score the highest for the presentations.

FRR Presentation Scoring Formula

The scoring of the Presentation is based on the average of the Presentation Judging forms. There is a maximum of 100 points on the Presentation Judging Form that will be scaled to meet the required portion of the competition total score.

Competition Flight

Launch and Flight Format

The launch will take place at a site determined by Tripoli Wisconsin Association. Each rocket must pass a safety inspection before launch, and the Range Safety Officer (RSO) must clear any additional equipment before entering the launch area. The RSO will have discretion over the number of TRA/NAR team members permitted to accompany the rocket once it is within the launch area. Each team must assemble a recovery team that will follow the directions of the RSO or designee.

To be considered a safe and successful flight, the rocket must:

- Safe Flight:
  - Launch followed by stable, near-vertical trajectory during ascent
  - Rocket's electronic, dual-deployment recovery system functions successfully, deploying drogue and main parachutes (and motor backup deploys successfully)
  - Rocket booster and dart must be recovered safely and in ready-to-fly condition
- Successful Flight:
  - Safe Flight
  - Apogee does not exceed the maximum altitude of 4,000 ft AGL ceiling
  - Acquisition and retention of downward-looking video from the dart throughout flight

A rocket will be considered in ready-to-fly condition if, after receiving another motor and resetting the deployment charges, it could pass RSO inspection, be returned to the launchpad, and safely fly again without repairs. The entire rocket must be returned to a designated location for the RSO or designee to conduct a post-flight review.

A flight performance report sheet (see Table 4 sample below) will be filled out by a designated flight operations recorder. The flight operations recorder will record the data on the sheet during and following the flight. Upon completion, a team member must sign their initials of acceptance before a copy will be released to the team.

Evaluation Criteria

Flights will be evaluated based on the maximum separation between the booster's and the dart's maximum altitudes, in addition to accuracy of predicted altitudes and separation and the successful recovery of downward-looking video.

Scoring Formula

There is a maximum of 100 points from the Competition Flight that will be scaled to meet the required portion of the competition total score. No less than 25 points will be awarded to rockets that complete a safe flight.

Post-Flight Launch Assessment Review (PLAR)

PLAR Report

The PLAR Report must be no more than eight (8) single-sided pages in length and must include, at a minimum:

- **Cover Page**
  - Report Title
  - WSGC CRL 2027
  - Team Name
  - School
  - Team Members' Names
- **Assessment of the rocket operation**
  - Pre-launch
  - Launch
  - Ascent
  - Deployment
  - Descent
  - Recovery
- **Performance of the payload**
  - Plots of altitude and acceleration values versus time
  - Discussion of the performance of the dart separation and how it might have been improved
    - Compare predicted and actual apogees; describe and defend possible reasons for differences
    - Discuss implemented separation system design, performance, and how it could have been improved for future flights
  - Actual vs. predicted flight performance
    - A "Flight Performance Reporting Sheet" (see Table 4 sample below)
    - Plot: "Acceleration Performance Comparison of Predicted and Actual for Rocket" (Figure 3 below)
    - Compare predicted and actual accelerations; describe and defend possible reasons for differences
    - Discussion of how flight could have been improved
  - Photographic documentation of the flight and team members

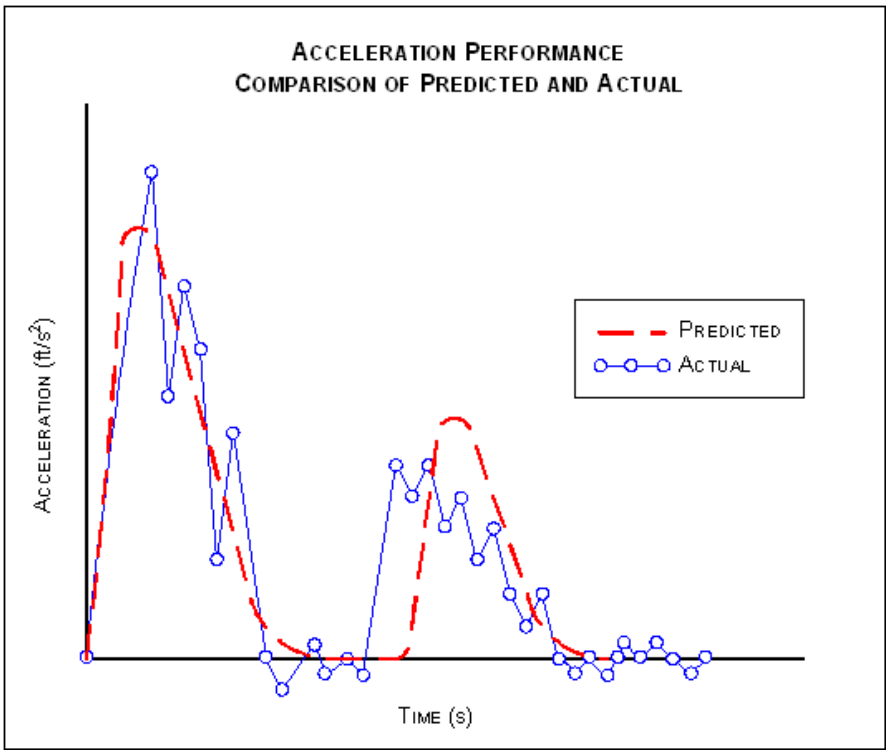


Figure 3: Example of Acceleration Performance Plot

SCHOOL _____ **Team** _____

1 Operation (determined by RSO or designee)	✓
Launch	
Stable Ascent	
Electronic, Dual-Deployment Recovery	
Drogue deployment	
Main parachute deployment	
Recovered	
Determined to be in flyable condition	

		Predicted (FRR)	**Actual**	
2 Maximum Altitude	*ft.(m)*			

Table 4: Example of Post-Launch Analysis Characteristics Table

PLAR Report Format

The performance comparison document should follow the same guidelines as the Design Report, be no more than eight (8) pages in length, and must be submitted electronically in *.pdf* format to the WSGC Grant Management page.

PLAR Report Evaluation Criteria

Reports will be evaluated on how closely the predicted results compare to the actual results, how well the team explains any differences, clarity, completeness, and professionalism of the material.

PLAR Report Scoring Formula

PLAR scoring is based on the average of the PLAR Report Judging forms. There is a maximum of 100 points from the Post-Launch Analysis Review Report Judging Form that will be scaled to meet the required portion of the competition total score.

Project Management

Project Management Performance

The ability of the team to meet the project targets will be assessed during this competition. Targets measured will include:

- Review and discussion of the team's design at five important phases with their rocketry mentor. Teams are required to submit the reports online by the deadlines posted in the competition calendar:
    - PDR – Preliminary Design Review
        - The PDR demonstrates that the preliminary design of all system requirements has been accounted for and that the proposed design approach has sufficient maturity to ensure success of the final design
    - CDR – Critical Design Review
        - The CDR demonstrates that the design is ready for fabrication. Systems are fully accounted for, and detailed hardware/software specifications can meet the functional requirements
    - FRR – Flight Readiness Review (Report)
        - The FRR incorporates the adjustments to the design given feedback from the Safety Review as well as any payload performance testing and rocket flight tests.
    - FRR – Flight Readiness Review (Presentation)
        - The FRR demonstrates that not only is the system ready to fly, but the procedures have been clearly defined for pre-flight preparation, the operation, and the post-flight safety and data download.
    - PLAR – Post-Launch Assessment Review
        - The PLAR evaluates the status, performance, and capabilities of the rocket evident from the flight operations experience since launch.
- Team representation at all meetings, either online or in person, as required during the competition.
- Completion of online forms to track the team's progress.
- Completion and quality of team documentation, in addition to the reports that have been spelled out. For example:
    - Team member and contact list, account profile setup, photos
    - Project budget
    - RockSim electronic files
    - Model rocket launch photos/videos

Evaluation Criteria

Timeliness, accuracy, quality, and thoroughness will be the basis for the evaluation.

Scoring Formula

Project Management Performance will be documented throughout the competition for each team. The result of the performance will be scaled to meet the required portion of the competition total score.

Educational Outreach

Educational Outreach Performance

The competition includes an Educational Outreach element in which each team shares aerospace-related information with a group. Outreach efforts will be evaluated based on the audience reached, the type and scope of activities conducted, and the extent of ongoing engagement. Consideration will be given to whether the event was new or continuing, its duration and format, and any follow-up efforts or documented impact.

Outreach possibilities could include, but are not limited to:

- Meet with a K-12 class or student organization to explain how rockets work.
- Make a presentation in the community or to a group on campus to describe the rocket competition and your team's design.
- Make a presentation to a group on campus describing opportunities at NASA or through the WSGC that are available to students before they graduate.

Evaluation Criteria

At the completion of the outreach event the team will need to have a representative at the event fill out and return to them an EPO (Education/Public Outreach) form that the team must then submit to the WSGC.

Scoring Formula

Teams that successfully complete the Educational Outreach and submit their EPO form will receive the portion of their total score based on audience reached, the type and scope of activities conducted, and the extent of ongoing engagement.

Appendix

Appendix A-1 PDR Report Evaluation Example

School _____ **Team** _____

Preliminary Design Review Report Evaluation

0.0	=	inadequate or no attempt
¼ Val	=	attempted but below expectation
½ Val	=	average or expected
¾ Val	=	above average but still lacking
Max Val	=	excellent, perfectly meets intent

_____ **DESIGN GOALS**

- Lists design goals for the rocket and payload
  - Describes how rocket mechanical performance measured relative to challenge
  - Describes how payload performance measured relative to challenge
  - Identifies strategy for design (focus on rocket, then payload, or simultaneous)
  - Identifies rationale for component deemed most important, other considerations

_____ **DESIGN CONSTRAINTS**

- Discusses of limitations on design
  - Limits imposed by the competition challenge
  - Limits imposed by the motor options
  - Limits team chose to include in addition to competition requirements
- Lists limits on resources
  - Budget, Personnel, Skills/Tools, Time

_____ **PRELIMINARY SOLUTION IDEA**

- Describes possible methods considered, decision process for solution chosen
- Mentions results of RockSim simulations, motor, dimensions, CP, CG
- Includes images or sketches of proposed rocket & payload, details
- Describes plan for avionics, on-board video, and recovery

_____ **PROJECT MANAGEMENT**

- Proposed Schedule
  - Identifies competition deliverable & meeting dates
  - Identifies school important dates (school breaks, final exam days, etc.)
  - Key design, fabrication, documentation, subsystem & system test milestones
- Proposed Budget
  - All rocket & payload components, travel expenses

_____ **SAFETY & MISCELLANEOUS**

- Safety
  - Designed for safe flight & recovery
  - Pre & post launch checklists
- Miscellaneous
  - Followed specifications, formatting, correct spelling & grammar
  - Documented figures and graphs, references and labeling

_____ **TOTAL = PDR REPORT POINTS**

COMMENTS:

Picture 1: Example of Preliminary Design Review (PDR) Report Evaluation Example

Appendix A-2 CDR Report Evaluation Example

School _____ **Team** _____

Critical Design Review Report Evaluation

0.0	=	inadequate or no attempt
¼ Val	=	attempted but below expectation
½ Val	=	average or expected
¾ Val	=	above average but still lacking
Max Val	=	excellent, perfectly meets intent

_____ **DESIGN FEATURES OF PAYLOAD SYSTEM**

- Possible boosted dart methods identified
  - Brief description of each possible method identified
  - Comparisons of each possible method
  - Describes decision process used for the evaluation and selection of top method
  - Design details of system to capture down-looking video from dart

_____ **DESIGN FEATURES OF RECOVERY SYSTEMS**

- Selected altimeter for recovery system deployment, operation and mounting details
- Dual-electronic deployment system, motor backup
- Parachutes selected, shock cords and mounting
  - Streamer acceptable in place of drogue on Dart
- Downed rocket location aid

_____ **ROCKET DIAGRAM & CONSTRUCTION**

- Accurate diagrams of rocket (CP, CG fully loaded on launchpad and at motor burnout)
  - External appearance with overall dimensions
  - Internal organization identifying components
- Construction solutions & techniques
- Structural analysis of custom-fabricated parts (if applicable)
- Photos of construction, rocket airframe, components, and payload

_____ **ANALYSIS OF ANTICIPATED PERFORMANCE**

- Prediction with description of how each prediction was made
  - Liftoff thrust to weight ratio
  - Launch rail guide velocity (speed at 10 ft altitude)
  - Maximum altitude of booster and dart, plot vs time (from start of flight)
  - Peak acceleration of booster and dart, plot vs time (from start of flight)

_____ **SAFETY and MISCELLANEOUS**

- Safety
  - Designed for safe flight & recovery
  - Planned pre & post launch procedures and checklists
- Miscellaneous
  - Followed specifications, formatting, correct spelling & grammar
  - Documented figures and graphs, references and labeling

_____ **TOTAL = CDR REPORT POINTS**

COMMENTS:

Picture 2: Example of Critical Design Review (CDR) Report Evaluation Example

Appendix A-3 FRR Report Evaluation Example

School _____ **Team** _____

Flight Readiness Review Report Evaluation

0.0	=	inadequate or no attempt
¼ Val	=	attempted but below expectation
½ Val	=	average or expected
¾ Val	=	above average but still lacking
Max Val	=	excellent, perfectly meets intent

_____ **RESULTS OF SYSTEM PERFORMANCE VERIFICATIONS**

- Avionics system, ground tests
- Onboard video system testing
- Downed rocket location system
- Test Flight, if performed

_____ **ADJUSTMENTS TO DESIGN OF ROCKET & PAYLOAD**

- Summarize final parameters, document any adjustments made, include photographs
  - Avionics system
  - Booster and/or dart
  - Onboard video system
  - Downed rocket location aid

_____ **ADJUSTMENTS TO DESIGN OF RECOVERY SYSTEM**

- Volume (in^3) of main deployment chamber(s)
- Grams of black powder and area (mm^2) of air holes in main deployment chamber(s)
- Number and size of sheer pins used for main deployment chamber(s)
- Parachute and shock cord configuration, altimeter settings

_____ **ADJUSTMENTS TO ANTICIPATED PERFORMANCE**

- Rocket Parameters – describe how each was determined
  - Dimensions, weight, aerodynamic drag, motor thrust to weight ratio
- Anticipated overall flight, maximum altitudes, peak accelerations, profiles vs time
- Stability analysis
  - CP and CG's identified, fully loaded and at motor burnout
- Environmental conditions analysis

_____ **SAFETY AND MISCELLANEOUS**

- Designed for safe flight & recovery
- Planned pre & post launch procedures and Checklists Followed Specifications
- Followed specifications, formatting, correct spelling & grammar
- Documented figures and graphs, references and labeling

_____ **TOTAL = FRR REPORT POINTS**

COMMENTS:

Picture 3: Example of Flight Readiness Review (FRR) Report Evaluation Example

Appendix A-4 PLAR Report Evaluation Example

School _____ **Team** _____

Post-Flight Launch Assessment Report Evaluation

0.0	=	inadequate or no attempt
¼ Val	=	attempted but below expectation
½ Val	=	average or expected
¾ Val	=	above average but still lacking
Max Val	=	excellent, perfectly meets intent

_____ **ASSESSMENT OF ROCKET OPERATION**

- Flight Anomalies Analysis

{If no anomaly then points are distributed across remaining subsections}

  - Propulsion System Assessment
  - Flight Path Assessment
  - Rocket Location & Recovery Analysis
  - Pre & Post Launch Procedure Assessment

_____ **PAYLOAD SYSTEM PERFORMANCE**

- Avionics system performance
- Separation of booster and dart
- Downward video from dart
- Discussion of overall payload performance and possible improvements

_____ **ACTUAL VS PREDICTED FLIGHT PERFORMANCE**

- Altitude comparison for booster
- Altitude comparison for dart
- Acceleration comparison for booster
- Acceleration comparison for dart

_____ **PHOTOGRAPHIC DOCUMENTATION OF FLIGHT**

- Launch pad and/or liftoff photo(s)
- In-flight photo(s)
- Landed on ground (or in tree, lake, etc.) before collected
- Recovered rocket showing all intact or damage

_____ **REPORT AESTHETICS**

- Followed specifications, formatting, correct spelling & grammar
- Professionally written
- Accurate representation of events, conclusions based on evidence and logic
- Documented figures and graphs, references and labeling

_____ **TOTAL = PLAR REPORT POINTS**

COMMENTS:

Picture 4: Example of Post-Flight Launch Assessment Review (PLAR) Report Evaluation Example

Appendix A-5 Outreach Form

Collegiate Rocket Launch
High-Power Rocket Competition
Hosted by Wisconsin Space Grant Consortium

Education/Public Outreach
Documentation Form

The Wisconsin Space Grant Consortium (WSGC) and NASA would like to thank you for giving our high-power rocket competition participants a chance to assist your organization. Please take a moment to fill in some information below to verify the students' participation. A portion of your team's competition score is based on their outreach activities. **Fill out one form for each outreach event you conduct**.

The goal of this activity is to "raise awareness of, or interest in, NASA, its goals, missions and/or programs, and to develop an appreciation for and exposure to science, technology, research and exploration." We are grateful for your involvement in this mission and we encourage you to be a part of additional projects that are taking place through NASA funding. If you have any questions about the competition or our organization, please visit our website at https://spacegrant.carthage.edu/

Your Team Information		
Team Lead's Name	**Advisor's Name**	**Academic Institution**
Team Lead Signature	**Advisor's Signature**	**Today's Date**

Event Information		
Date of Event	**Approximate # of Attendees**	**List Each City, State, & Zip Code Where the Event Took Place**
Brief Description of Attendees (Circle all that apply)		**List All Organizations Involved With the Event**
PreK: Students Teachers K-5 Grade: Students Teachers 6-8th Grade: Students Teachers High School: Students Teachers University/ Campus: Undergrads Graduate Administrator Faculty Public at Large Informal Education Setting (Museum, etc.)		

Picture 5: Education Outreach Form, page 1 of 2

Event Information Continued	
Brief Description of Activity	
Is this a new or existing event? (Circle one)	**What was the duration of the event? (Circle one)**
New Existing	< 2 days = 2 days > 2 days
How many exhibits were supported/developed by this event?	**How many student hands-on activities were supported/developed by this event?**
How many public at large activities were supported by this event?	**If other activities were supported by this event, please explain:**
Please provide links to any media coverage (via your institution, local, or regional news outlets) received for this event:	
Please provide the title, presenter, and venue for any presentations directly attributed to this activity.	
Describe how your team plans to build upon this outreach event:	
Please use this space to provide WSGC with any additional information about this outreach event:	

Picture 6: Education Outreach Form, page 2 of 2